ROC Curve and AUC

Evaluating Classification Models

Why Do We Need ROC and AUC?

Suppose a classification model predicts whether an email is spam.
The model does not always directly produce only:

Spam or **Not Spam**

Instead, many classifiers produce a probability or score.
For example:

Email	Spam Probability
E1	0.95
E2	0.82
E3	0.61
E4	0.48
E5	0.20

We choose a **threshold** to convert the score into a class.

What Happens When We Change the Threshold?

Consider the same model.

Threshold	Effect
0.90	Very few samples predicted Positive
0.70	More samples predicted Positive
0.50	More samples predicted Positive
0.30	Even more samples predicted Positive
0.10	Almost everything predicted Positive

Changing the threshold changes:

- True Positives
- False Positives
- True Negatives
- False Negatives

Therefore, Precision and Recall also change.

Two Important Quantities

ROC uses two quantities:

True Positive Rate (TPR)

$$\boxed{TPR = \frac{TP}{TP + FN}}$$

TPR is also called **Recall** or **Sensitivity**.

False Positive Rate (FPR)

$$\boxed{FPR = \frac{FP}{FP + TN}}$$

Therefore:

ROC Curve = TPR plotted against FPR

Understanding TPR and FPR

Suppose we have:

$$TP = 40, \quad FN = 10, \quad FP = 15, \quad TN = 35$$

Then:

$$TPR = \frac{40}{40 + 10} = \frac{40}{50} = 0.80$$

Therefore:

$$\boxed{TPR = 80\%}$$

For FPR:

$$FPR = \frac{15}{15 + 35} = \frac{15}{50} = 0.30$$

Therefore:

Building the ROC Curve

For different threshold values, calculate:

$$TPR = \frac{TP}{TP + FN}$$

and

$$FPR = \frac{FP}{FP + TN}$$

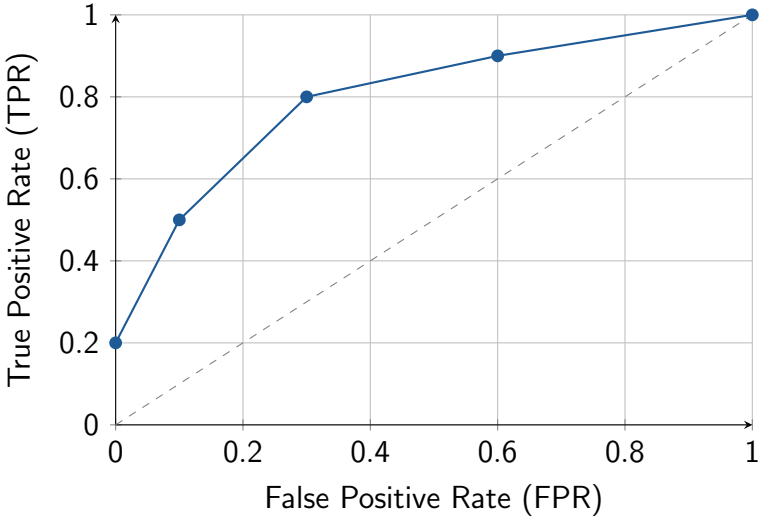
Then plot:

x-axis $\rightarrow$ FPR
y-axis $\rightarrow$ TPR

For example:

Threshold	FPR	TPR
0.90	0.00	0.20

ROC Curve



Why Is the Top-Left Corner Important?

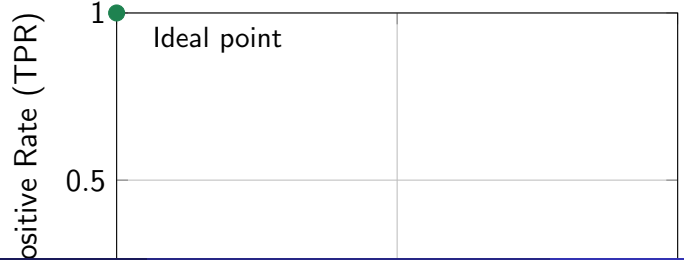
The ideal point is:

$$\boxed{FPR = 0, \qquad TPR = 1}$$

This means:

- No false positives.
- All actual positive cases are detected.

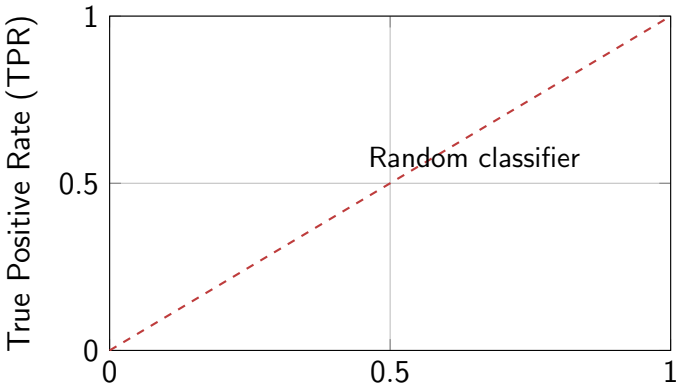
Therefore, the top-left corner represents an ideal classifier.



Random Classifier

A random classifier has no useful ability to distinguish between the two classes.
Its ROC curve is approximately the diagonal:

$$TPR \approx FPR$$



What Is AUC?

AUC stands for:

Area Under the ROC Curve

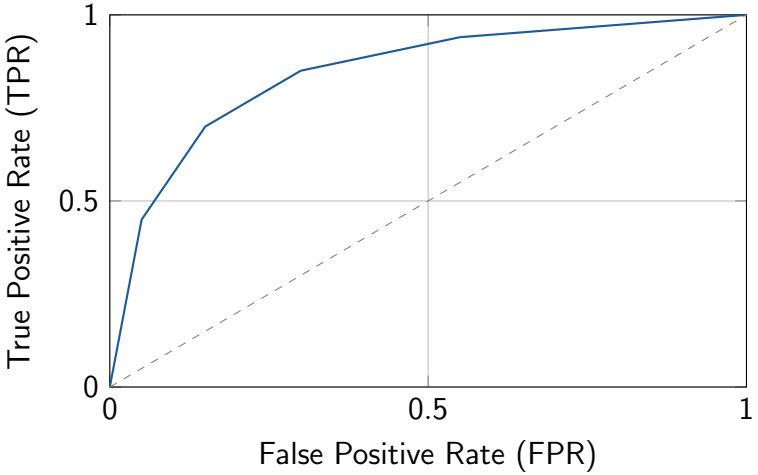
AUC summarizes the ROC curve into a single number.

$$\boxed{0 \leq AUC \leq 1}$$

Higher AUC means that the model generally does a better job of separating positive and negative samples.

AUC	General Interpretation
1.0	Perfect separation
0.8–0.9	Strong separation
0.6–0.8	Some separation
0.5	Approximately random

Visual Meaning of AUC



Key Idea

Simple Example: Two Models

Suppose two models are trained to detect fraudulent transactions.
Both models can be evaluated at many different thresholds.

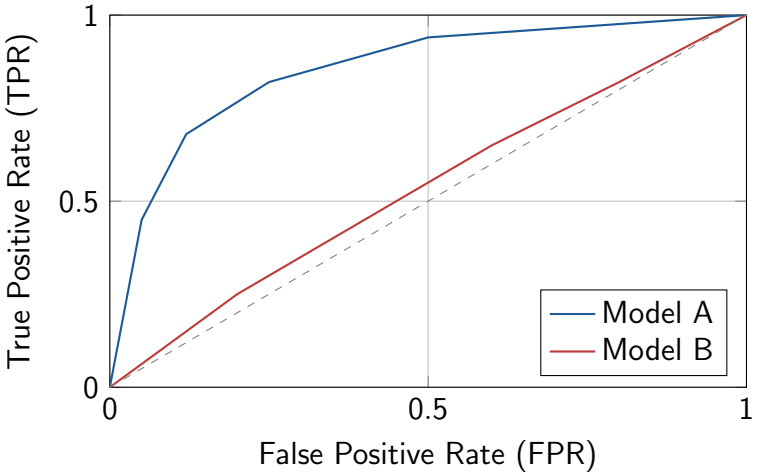
Model	AUC
Model A	0.92
Model B	0.68

The ROC curves summarize how TPR and FPR change across thresholds.

Interpretation

The AUC values describe the models' overall class-separation ability across thresholds.
Model A has the larger measured AUC in this example.

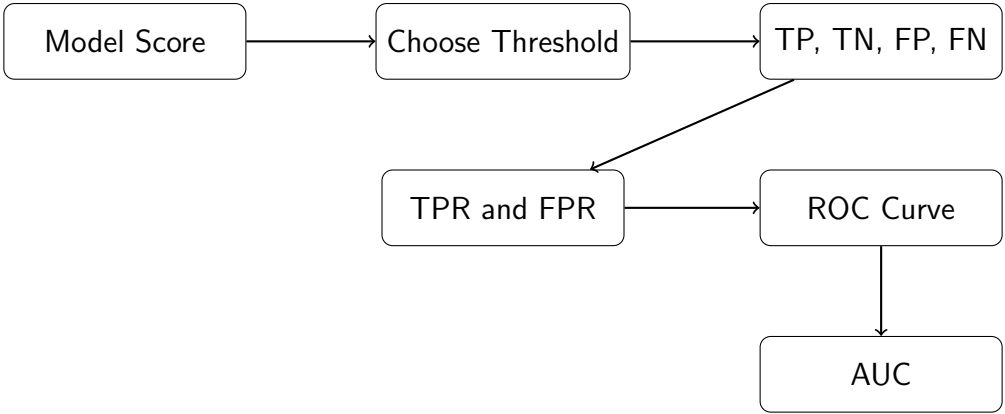
ROC Curves of Two Models



ROC vs AUC

Term	Meaning
ROC Curve	A graph showing the relationship between TPR and FPR at different classif
AUC	A single numerical value representing the area under the ROC c
ROC	Helps visualize the model's behavior across thresholds.
AUC	Provides a compact summary of the ROC curve.

ROC and Threshold: The Complete Picture



Remember

Different thresholds $\rightarrow$ different TPR/FPR $\rightarrow$ ROC curve $\rightarrow$ AUC

Important Points to Remember

- ROC stands for **Receiver Operating Characteristic**.
- ROC plots:

$$TPR \text{ against } FPR$$

- TPR is the same as Recall:

$$TPR = \frac{TP}{TP + FN}$$

- FPR is:

$$FPR = \frac{FP}{FP + TN}$$

- ROC is obtained by changing the classification threshold.
- AUC is the area under the ROC curve.
- AUC close to 1 indicates strong class separation.
- AUC around 0.5 corresponds to approximately random discrimination.

One-Line Memory Trick

ROC

FPR on X-axis TPR on Y-axis

AUC

Area Under the ROC Curve